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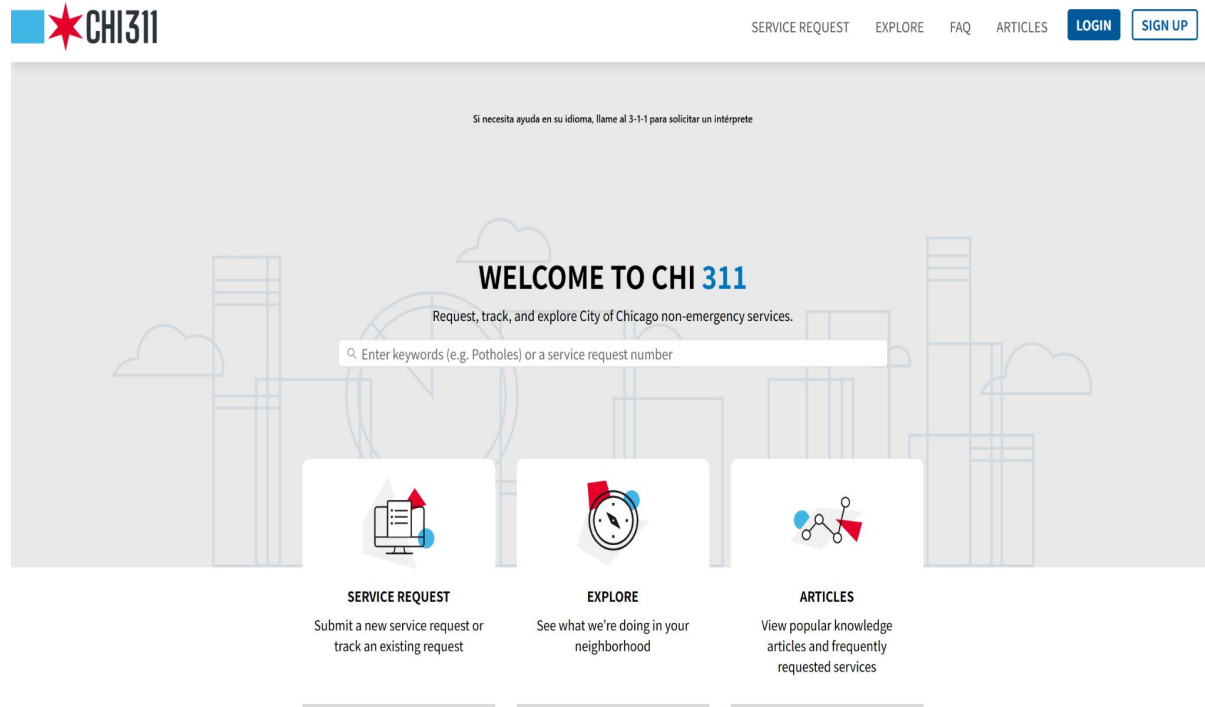
Chi 311 Agentic AI

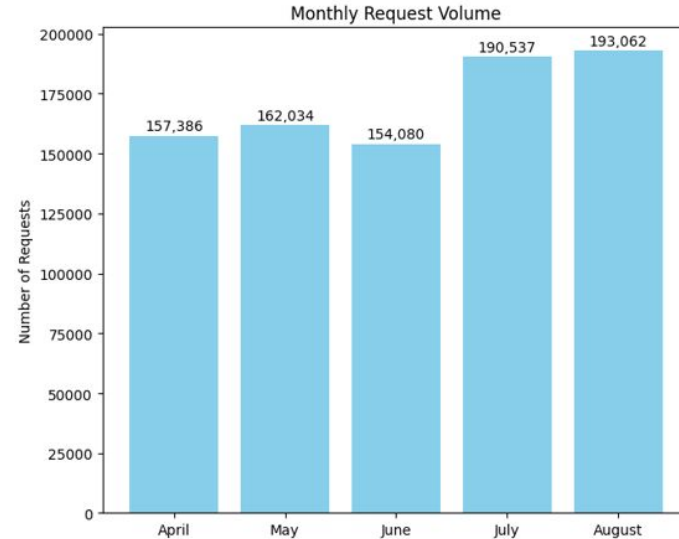
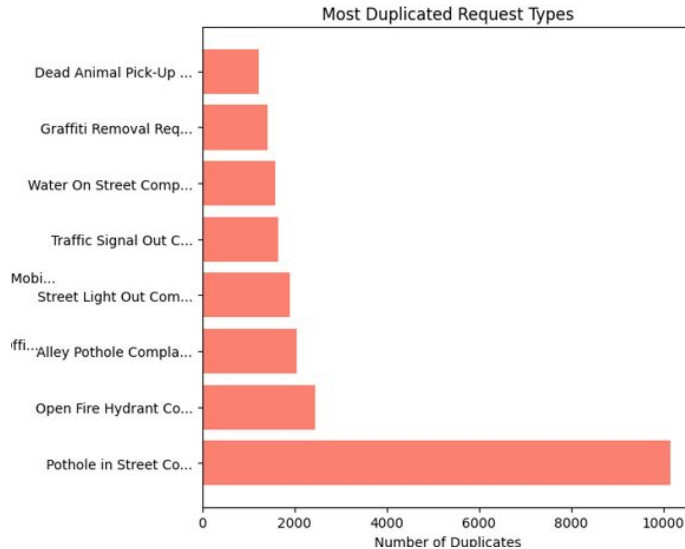
Talk, Snap, Report: Automating Civic Issue Reporting with AI Agent

Jung In Chang
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What is Chi 311?

- Civic service portal for Chicago residents to report non-emergency requests.
- Potholes, Graffiti, Airplane Noise complaints, Fraud, Animal rescues etc.
There are **MANY!**





**Given the scale of 311 reports,
Improving Chicago's 311 system can deliver meaningful, city-wide benefits.**

Submitting a 311 Request: Slow, Manual, and Complex


ANIMALS


CONSUMER AND
EMPLOYEE
PROTECTION


COVID-19 ASSISTANCE


DISABILITIES


GARBAGE AND
RECYCLING


HEALTH


HOME AND BUILDINGS


PARKS, TREES,
ENVIRONMENT


TRANSPORTATION
AND STREETS

Where

What

Who

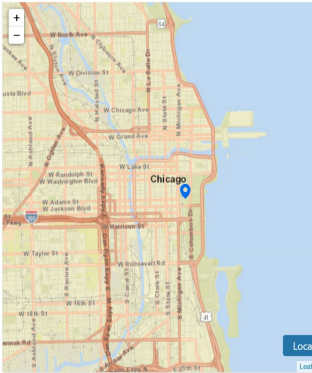
Summary

Where? (Address of Service Location)

Please provide the address of the request or complaint below. Select the address from the drop down option. Confirm the address prior to submission.

CANCEL

The CHI311 service request system relies on a database of city addresses maintained by the City of Chicago.



Locate Me

What?

Who

Summary

What?

Please answer the following questions so the city can receive your request.

Upload Images or Drag Files in

Upload Files

Or drop files

No File Selected

*1. What is the nature of this request?

2. What is the caller's relationship to Senior?

3. What is the Senior's LAST NAME?

4. What is the Senior's FIRST NAME?

5. What is the Senior's Phone Number?

6. What is the Senior's Age?

7. Is the Senior male or female?

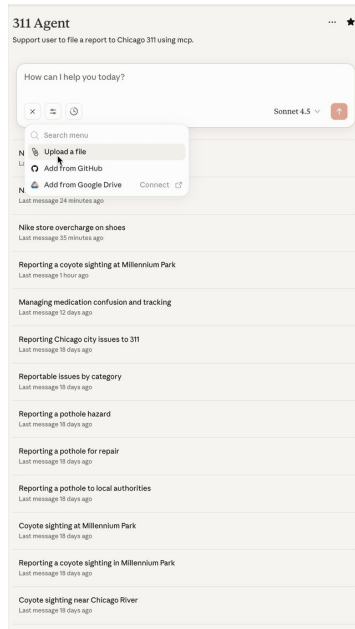
8. What language does the Senior speak?

*9. Is this a CHA property? Yes or No

CANCEL

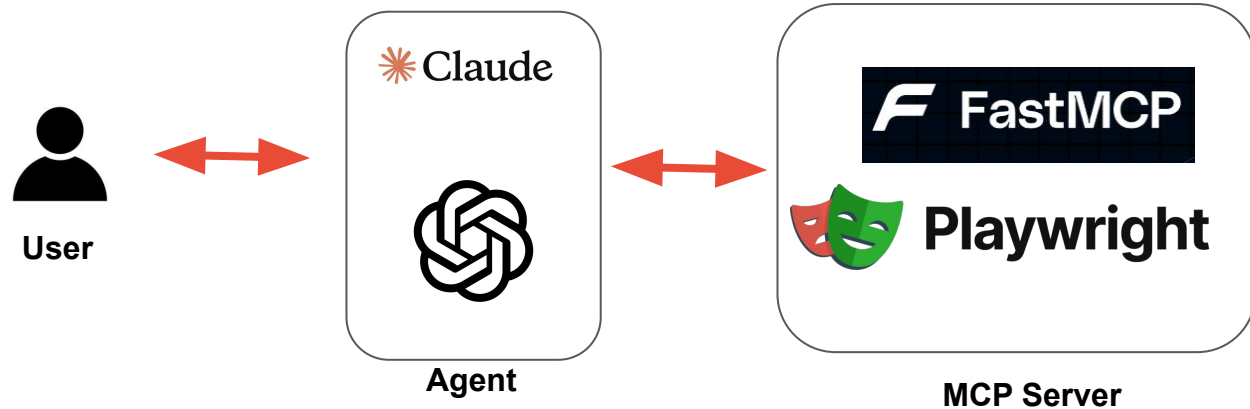
NEXT

Agent-based Auto Reporting



Architecture: Agent + Custom MCP Server + Prompting Strategy

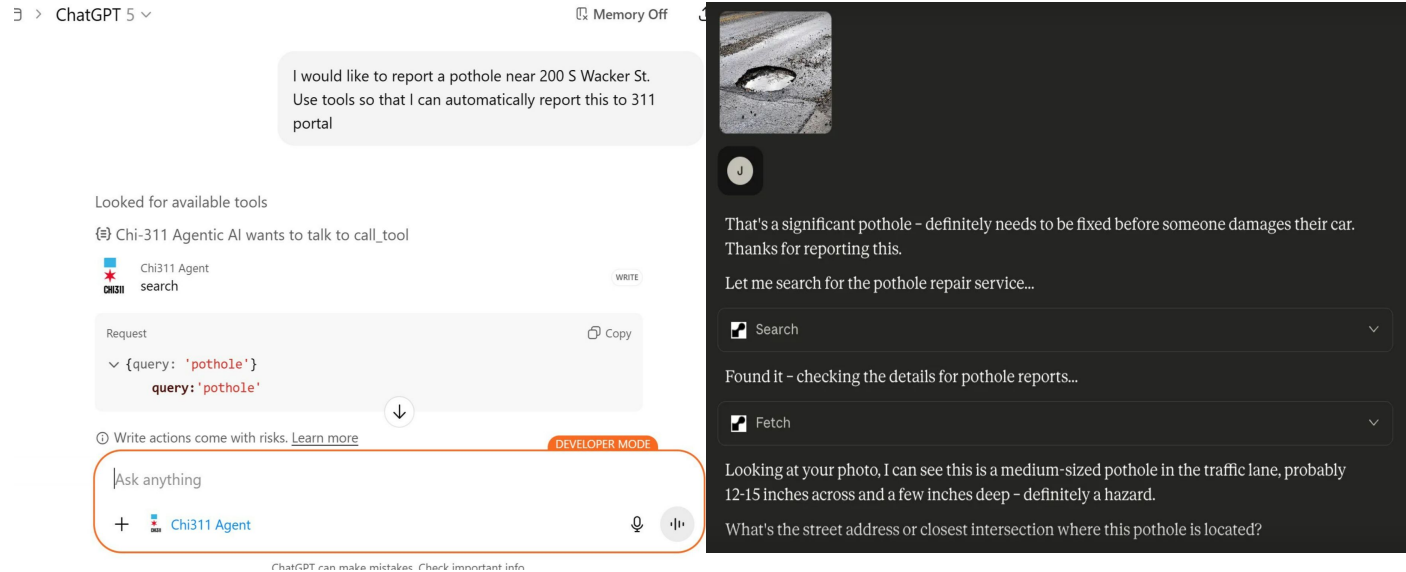
1. User describes the situation
2. Agent converts it into structured fields
3. MCP server fills & submits the actual 311 form



How MCP Submits a Real 311 Request

No 311 API
exists →
automated
browser using
Playwright





The screenshot displays the ChatGPT interface with a user request and an agent response. The user's message is: "I would like to report a pothole near 200 S Wacker St. Use tools so that I can automatically report this to 311 portal". The agent's response includes a photo of a pothole, a text analysis, and a search for repair services.

ChatGPT 5 ▾ Memory Off

I would like to report a pothole near 200 S Wacker St. Use tools so that I can automatically report this to 311 portal

Looked for available tools

Chi-311 Agentic AI wants to talk to call_tool

Chi311 Agent search

Request

```
{query: 'pothole'}  
query: 'pothole'
```

Write actions come with risks. [Learn more](#) **DEVELOPER MODE**

Ask anything

Chi311 Agent

Chi311 Agent

That's a significant pothole - definitely needs to be fixed before someone damages their car. Thanks for reporting this.

Let me search for the pothole repair service...

Search

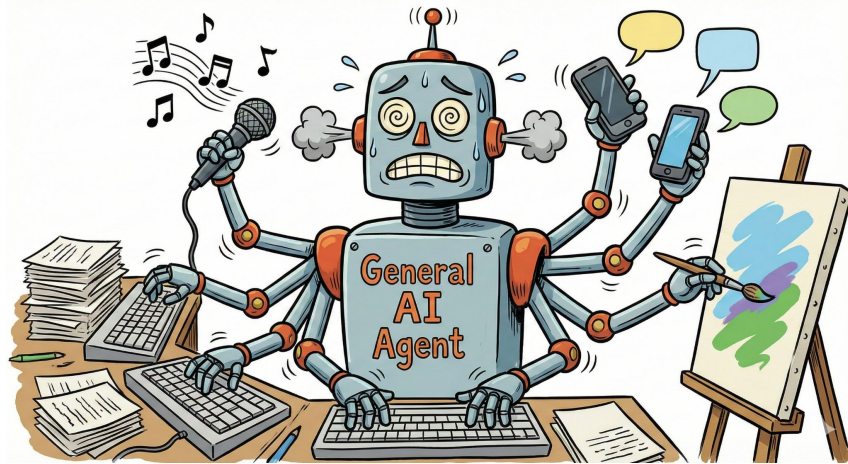
Found it - checking the details for pothole reports...

Fetch

Looking at your photo, I can see this is a medium-sized pothole in the traffic lane, probably 12-15 inches across and a few inches deep - definitely a hazard.

What's the street address or closest intersection where this pothole is located?

- We leveraged **Claude's** agentic behavior
- Chi311 can be added to **YOUR** Claude & ChatGPT
- **Supports Image & Audio & Web searching & ETC**



Without Prompt

- Handles general requests
- Slower to recognize 311 reporting intent



With Prompt

- Focuses on 311 reporting
- More concise, task-oriented, and friendly

Demo Scenario 1:

Retail Fraud Complaint (no prompt)

 **User**

 *"Hi, I was charged one hundred dollars for shoes marked at fifty at a Nike store today."*

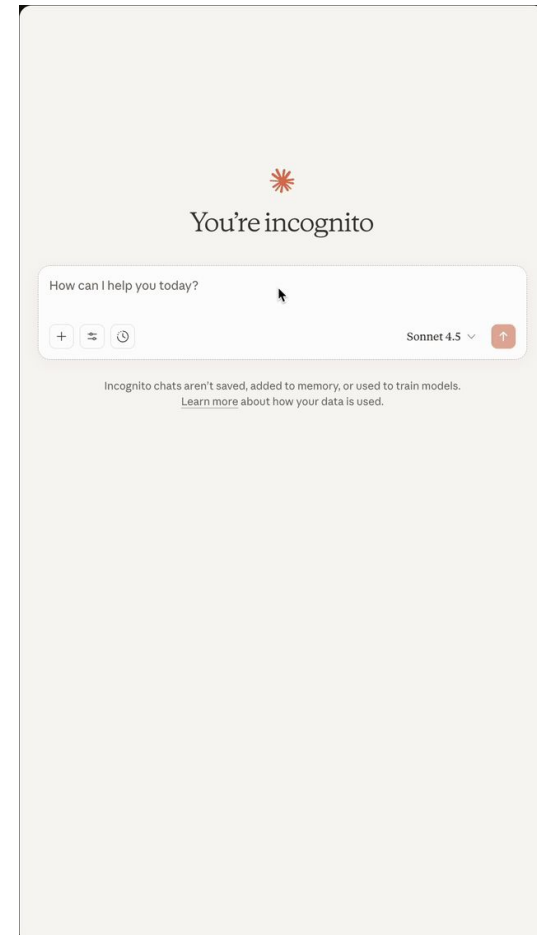
What Happens Without Prompting

 *"I'm sorry you were scammed — here's general advice on reporting fraud..."*

 *"You should contact your bank or the business directly."*

 *"I can explain how fraud reports usually work..."*

 *"Could you share more details about the incident?"*



Demo Scenario 1:

Retail Fraud Complaint (with prompt)

◆ What Happens *With* Prompting

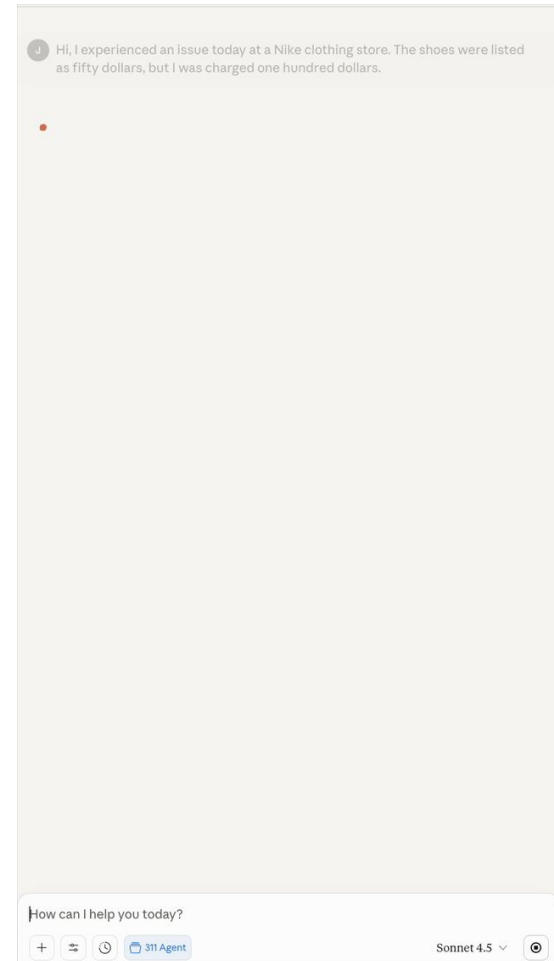
■ *"Identified this as a **Retail Fraud** 311 category."*

■ *"Checking if the business exists at 200 S Wacker Street..."*

■ *"No matching Nike store found — validating location by search."*

■ *"Extracted required 311 fields from your description."*

■ *"Preparing 311 Fraud Report with structured details..."*



Demo Scenario 2: Coyote Complaint (no prompt)

 **User**

 [A Photo of a Coyote @ Chicago Bean]

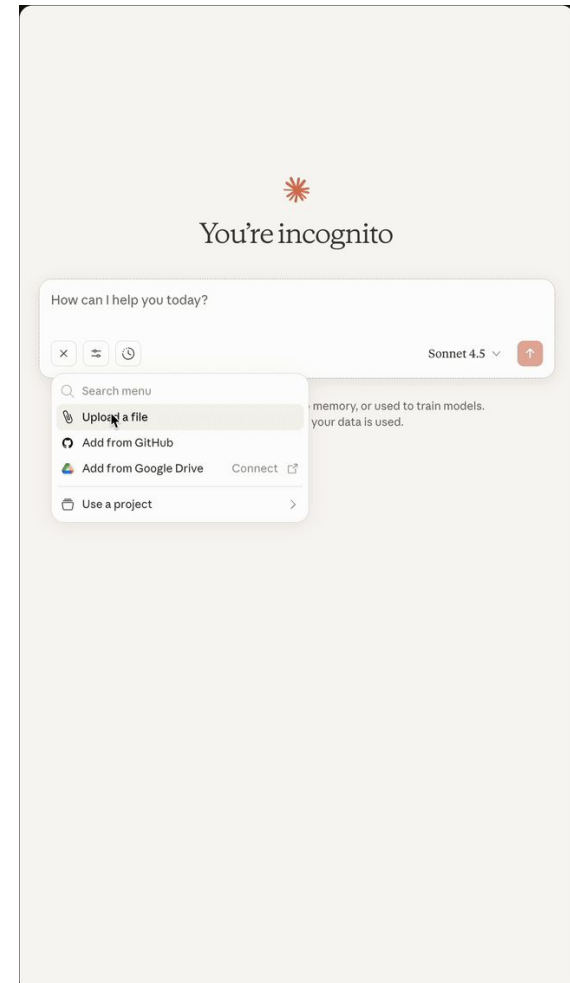
◆ What Happens Without Prompting

 “This appears to be an image of an animal in a park...”

 “It looks like a coyote standing on grass near a walkway.”

 “Let me know how I can help with this image.”

 “If you want to report something, please tell me what action to take.”



Demo Scenario 2:

Coyote Complaint (with prompt)

What Happens *With Prompting*

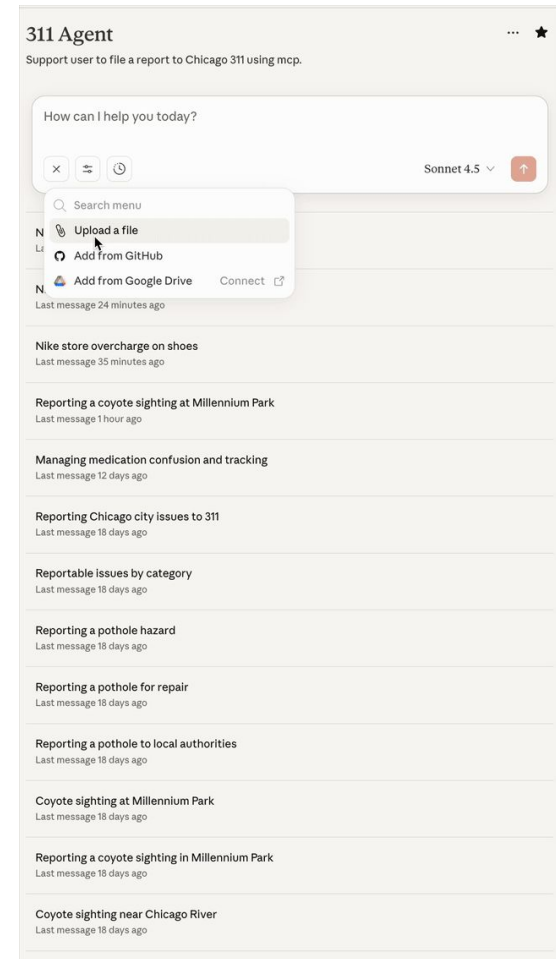
“Detected a **coyote sighting** at Millennium Park from your image.”

“Identified this as a **311 Animal Control** → **Coyote Complaint** case.”

“Extracted key fields: location, time seen, and animal behavior.”

“Preparing structured 311 report with minimal additional input...”

“Ready to submit the official request.”



Methods	Scenario 1 <i>Time</i>	Scenario 1 <i>Success</i>	Scenario 2 <i>Time</i>	Scenario 2 <i>Success</i>
Orig Chi 311	3 min		3 min	
Default Claude Agent	N/A		N/A	
Default Claude Agent (No Prompt) + MCP	4 min		2 min	
Claude Agent (With Prompt) + MCP	2 min		1 min	

- Scenario 1: Nike Fraud Complaint (**Text-based**)
- Scenario 2: Coyote @ Bean (**Image Only**)

Key Findings

- General LLMs can be extended into **specialized 311 agents** using MCP.
- **Prompting is essential**

Future Improvements

- Add **audio-based reporting** (e.g. aircraft noise)
- Use **image metadata (GPS/time)**

Conclusion

- Our system proves that **agentic AI + good prompting** can create practical **civic-tech** tools.
- This 311 agent can make city services **faster, easier, and more accessible** for Chicago residents.



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Thank you!

Question, Comments, Critics?

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